

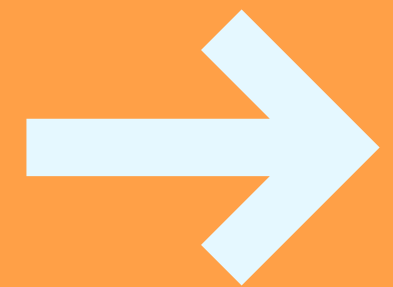
Species Distribution Modeling in the UK using presence-only data



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Abstract

Conservation challenge in the UK.
Role of Species Distribution Modeling (SDM) with presence-only data.

Five focal species.
Key methods: MaxEnt, DeepMaxEnt, ensembles.

Outcomes: Habitat suitability maps, identification of strongholds, policy contribution.

Problem Statement

Lack of reliable absence data

Sampling and observer bias issues.
Need for presence-only modeling innovations.

Goal: bridge computational modeling with conservation policy

Literature Review

Common SDM algorithms: MaxEnt,
BIOCLIM, DOMAIN, DeepMaxEnt

Bias handling: Target Group
Background (TGB).
Ensemble modeling improves reliability.

Validation metrics: AUC-ROC, Boyce Index
limitations and strengths

Research Questions

01

Where are the most environmentally suitable areas for the endangered species selected in the UK?

02

What are the key environmental conditions that conservation efforts should prioritize for the selected endangered species in the UK?

03

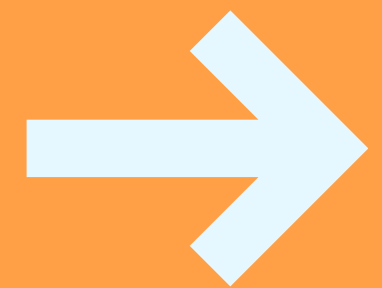
Implicit question: To what extent can advanced modeling techniques mitigate the data limitations inherent in presence-only datasets where true absences are unavailable?



Methodology Overview

A Comprehensive Approach to Species Selection and Data Gathering

Workflow: species selection → data collection →
cleaning/preprocessing → bias mitigation → modeling → evaluation
→ mapping.



Species Selection & Data Sources

Species: Hedgehog, Red Squirrel, Hazel Dormouse, Brown Hare, Soprano Pipistrelle Bat.

Data: GBIF for species occurrences, UKCEH for land cover, MODIS for Land Surface Temperature, Sentinel-2 for spectral indices.

Data Cleaning & Feature Engineering

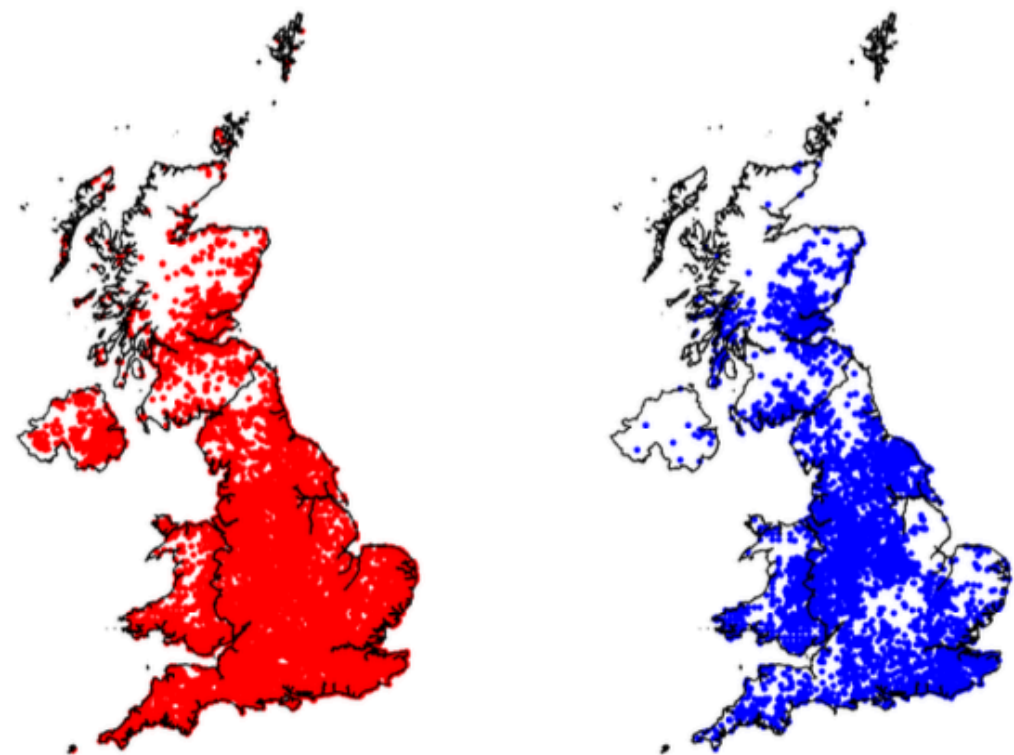
- Removal of duplicates/nulls.
- Proximity-based predator-prey features.
- One-hot encoding for land cover.
- Distance features for roads, wind turbines.

Bias Handling

- Sampling unevenness → false absences.
- Target Group Background points mitigate bias.

Presence Points

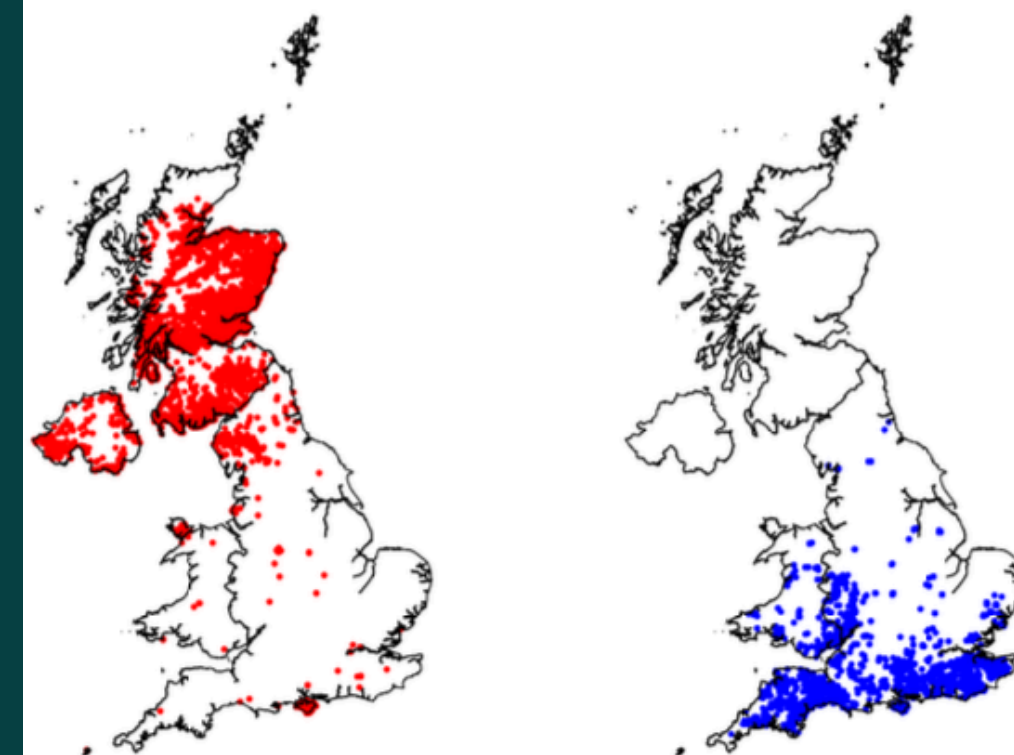
Background Points



(a) *Erinaceus europaeus*

Presence Points

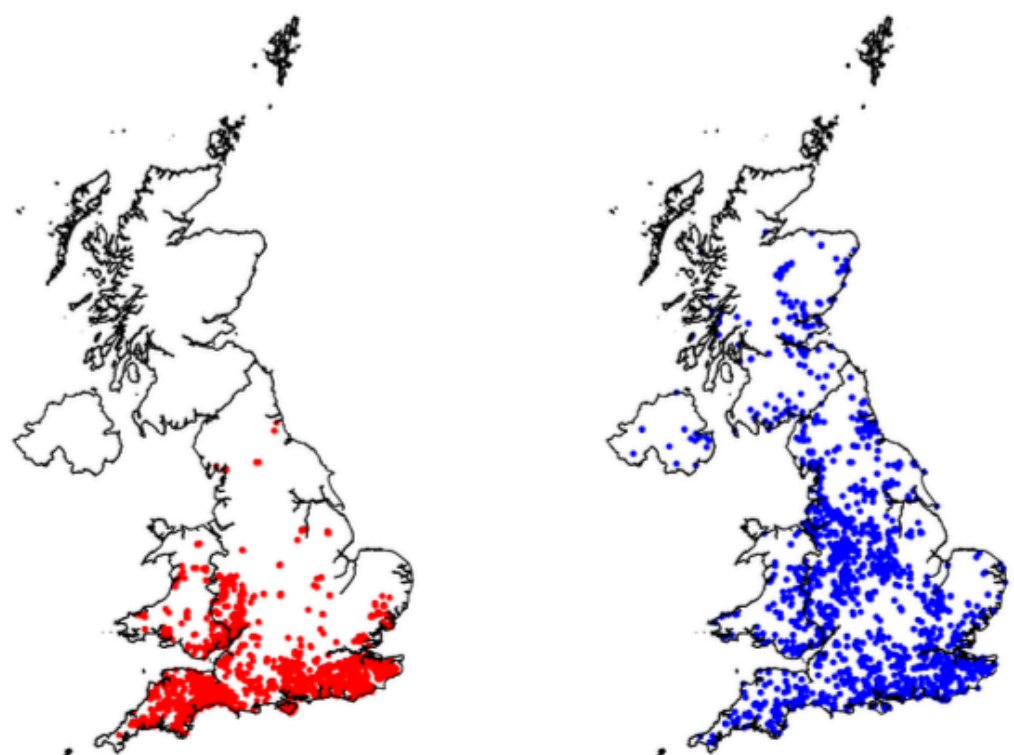
Background Points



(b) *Sciurus vulgaris*

Presence Points

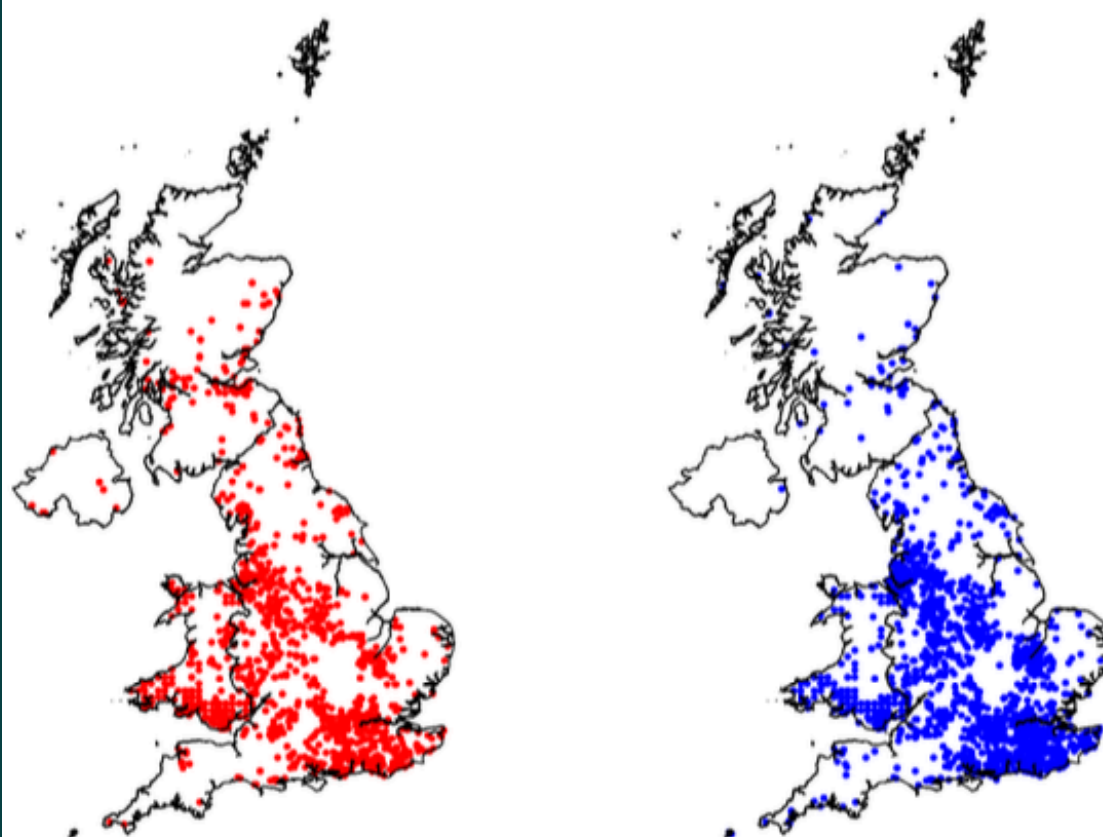
Background Points



(c) *Muscardinus avellanarius*

Presence Points

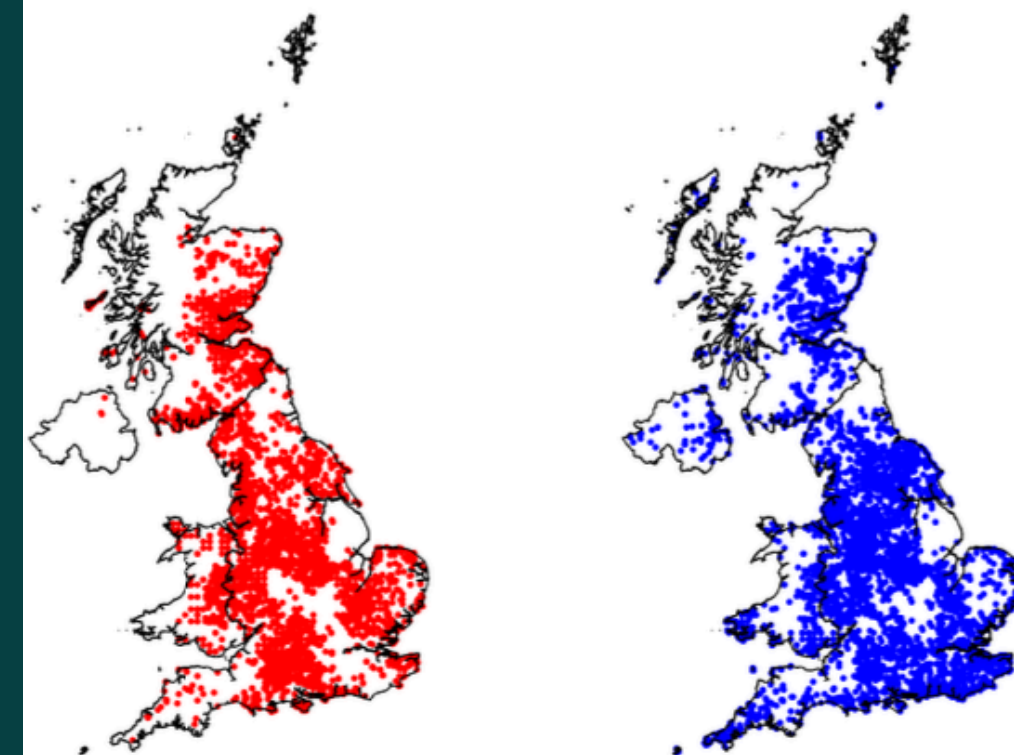
Background Points



(e) *Pipistrellus pygmaeus*

Presence Points

Background Points



(d) *Lepus europaeus*

Features influencing the Hedgehog

Environmental Factor	Influence	Details
Predator distribution (<i>Meles meles</i> (Badger))	-ve	Inverse relationship with <i>Erinaceus europaeus</i> occurrence.
Prey distribution (Ground beetles, e.g. <i>Pterostichus madidus</i> , <i>Harpalus</i> , <i>Nebria</i> , <i>Amara</i> , <i>Bembidion</i>)	+ve	Source of food.
Road proximity	-ve	Road casualties negatively impact local hedgehog populations.
Target Group Background species	-ve	<i>Talpa europaea</i> (European Mole), <i>Sorex araneus</i> (Common Shrew), and <i>Apodemus sylvaticus</i> (Wood Mouse) chosen as background species for modelling.

Features influencing the Red Squirrel

Environmental Factor	Influence	Details
Presence of Sitka spruce	-ve	Unfavourable habitat.
Competition with Grey squirrels	-ve	Grey squirrels compete with red squirrels for resources.
Target Group Back-ground species	-ve	<i>Tamias sibiricus</i> (Siberian Chipmunk), <i>Glis glis</i> (Edible Dormouse), and <i>Muscardinus avellanarius</i> (Hazel Dormouse).

Features influencing the Hazel Dormouse

Environmental Factor	Influence	Details
Presence of Hazel, Beech, Birk, Honeysuckle, Oak, and Hawthorn	+ve	These plant species provide food and nesting material.
Target Group Background species	-ve	<i>Eliomys quercinus</i> (Garden Dormouse), <i>Apodemus sylvaticus</i> (Wood Mouse), and <i>Apodemus flavicollis</i> (Yellow-necked Mouse).

Features influencing the Brown hare

Environmental Factor			Influence	Details
Winter wheat			-ve	Switch towards winter wheat may reduce food and shelter opportunities.
Target	Group	Back-ground species	-ve	<i>Lepus timidus</i> (Mountain Hare) and <i>Oryctolagus cuniculus</i> (European Rabbit).

Features influencing the Soprano Pipistrelle Bat

Environmental Factor	Influence	Details
Prey availability (Ceratopogonidae, Chironomidae)	+ve	These insect families provide food.
Wind turbine proximity	-ve	Wind turbines cause mortality in bats.
Road proximity	-ve	Vehicle collisions are a significant threat.
<i>Felis catus</i> (Cat) presence	-ve	Domestic cats frequently injure or kill these bats.
Target Group Background species	-ve	<i>Pipistrellus pipistrellus</i> (Common Pipistrelle) and <i>Pipistrellus nathusii</i> (Nathusius' Pipistrelle).

Modeling

MaxEnt: traditional presence-only model.

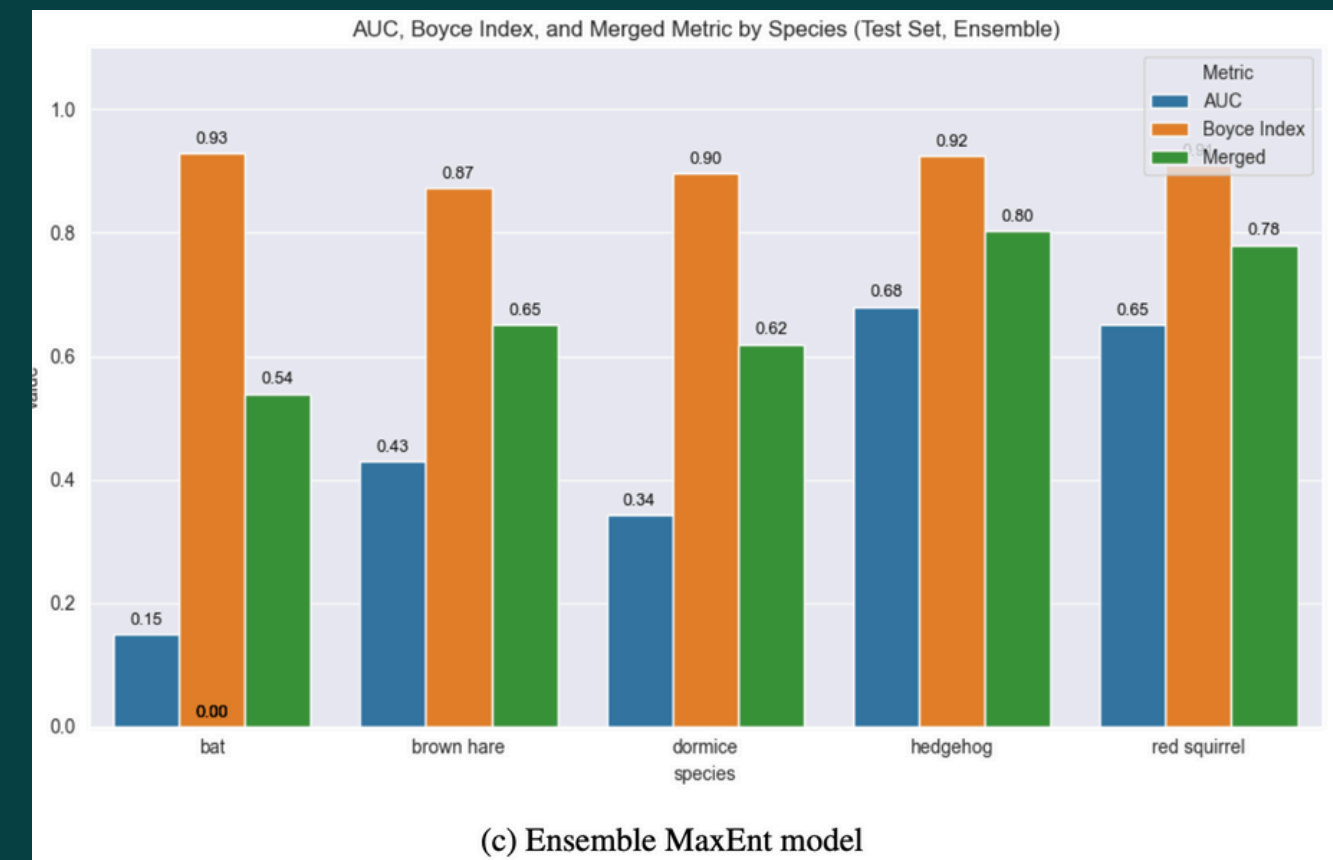
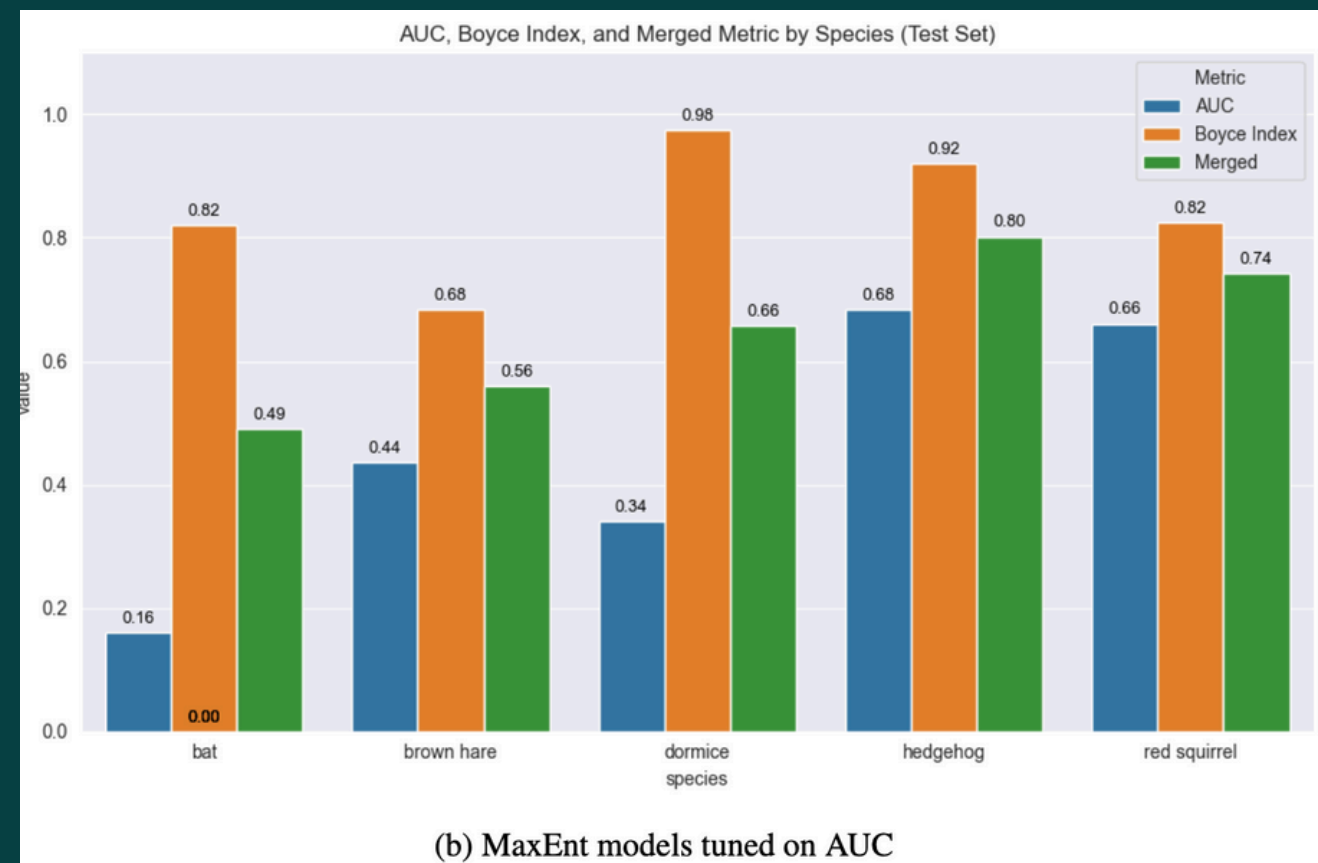
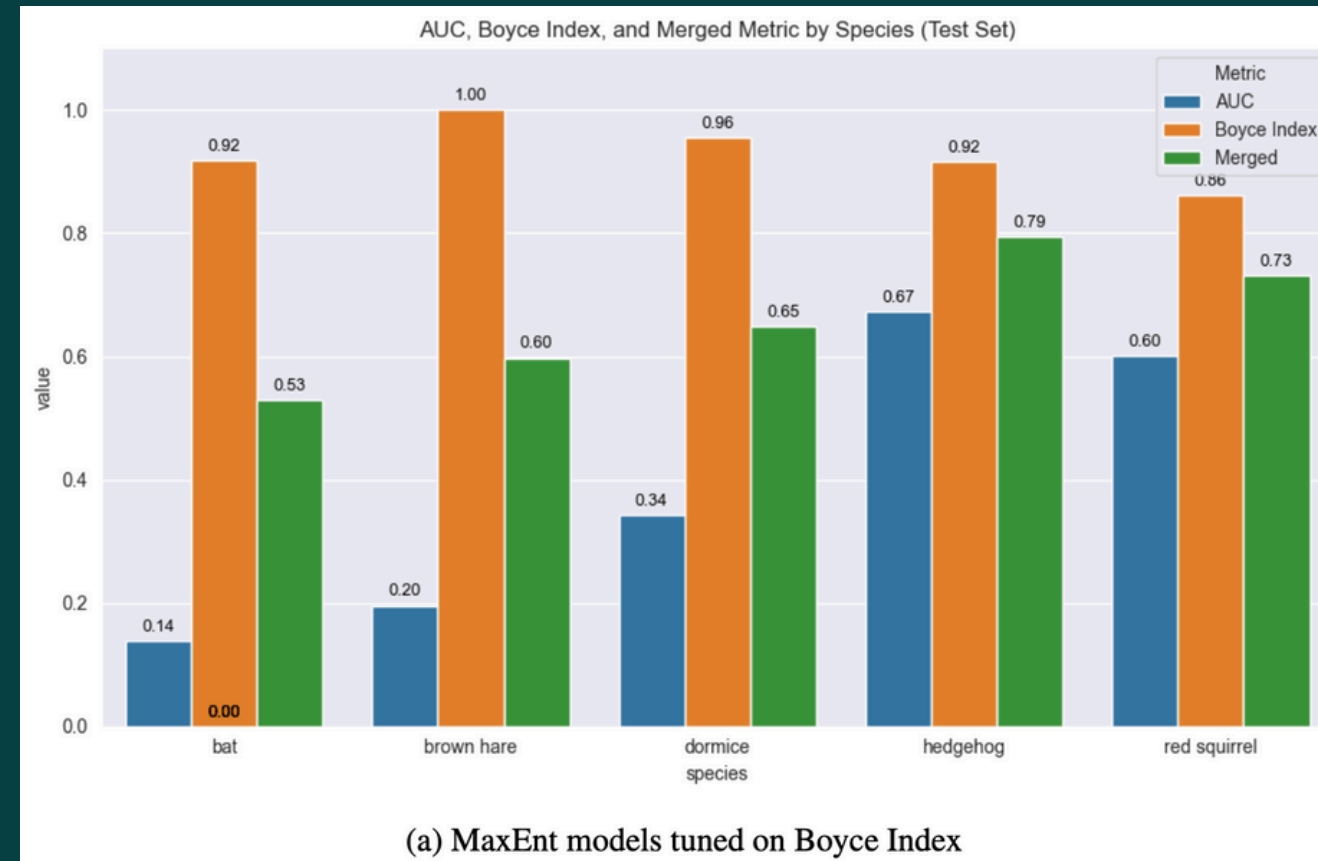
DeepMaxEnt: neural network-based, better under bias.

Ensembles: soft voting for robustness

Evaluation: AUC-ROC + Boyce Index merged metric

MaxEnt Performance

- Trade-offs between AUC and Boyce tuning
- Overfitting issues
- Ensembles improved performance



DeepMaxEnt Performance

- DeepMaxEnt consistently better
- Pipistrelle anomaly: MaxEnt inflated by Boyce Index, AUC too low → DeepMaxEnt chosen.

Species	Model(s)	Score
<i>Erinaceus europaeus</i> (Hedgehog)	BCEWithLogitsLoss tuned on Boyce Index + DeepMaxEnt Loss tuned on Boyce Index	0.810
<i>Sciurus vulgaris</i> (Red Squirrel)	DeepMaxEnt Loss tuned on Boyce Index + DeepMaxEnt Loss tuned on AUC + BCEWithLogitsLoss tuned on AUC + BCEWithLogitsLoss tuned on Boyce Index	0.798
<i>Muscardinus avellanarius</i> (Hazel Dormouse)	DeepMaxEnt Loss tuned on Boyce Index + DeepMaxEnt Loss tuned on AUC + BCEWithLogitsLoss tuned on AUC + BCEWithLogitsLoss tuned on Boyce Index	0.756
<i>Lepus europaeus</i> (Brown Hare)	BCEWithLogitsLoss tuned on Boyce Index + BCEWithLogitsLoss tuned on AUC + DeepMaxEnt Loss tuned on AUC	0.747
<i>Pipistrellus pygmaeus</i> (Soprano Pipistrelle Bat)	BCEWithLogitsLoss tuned on AUC	0.378

Visualization & Mapping

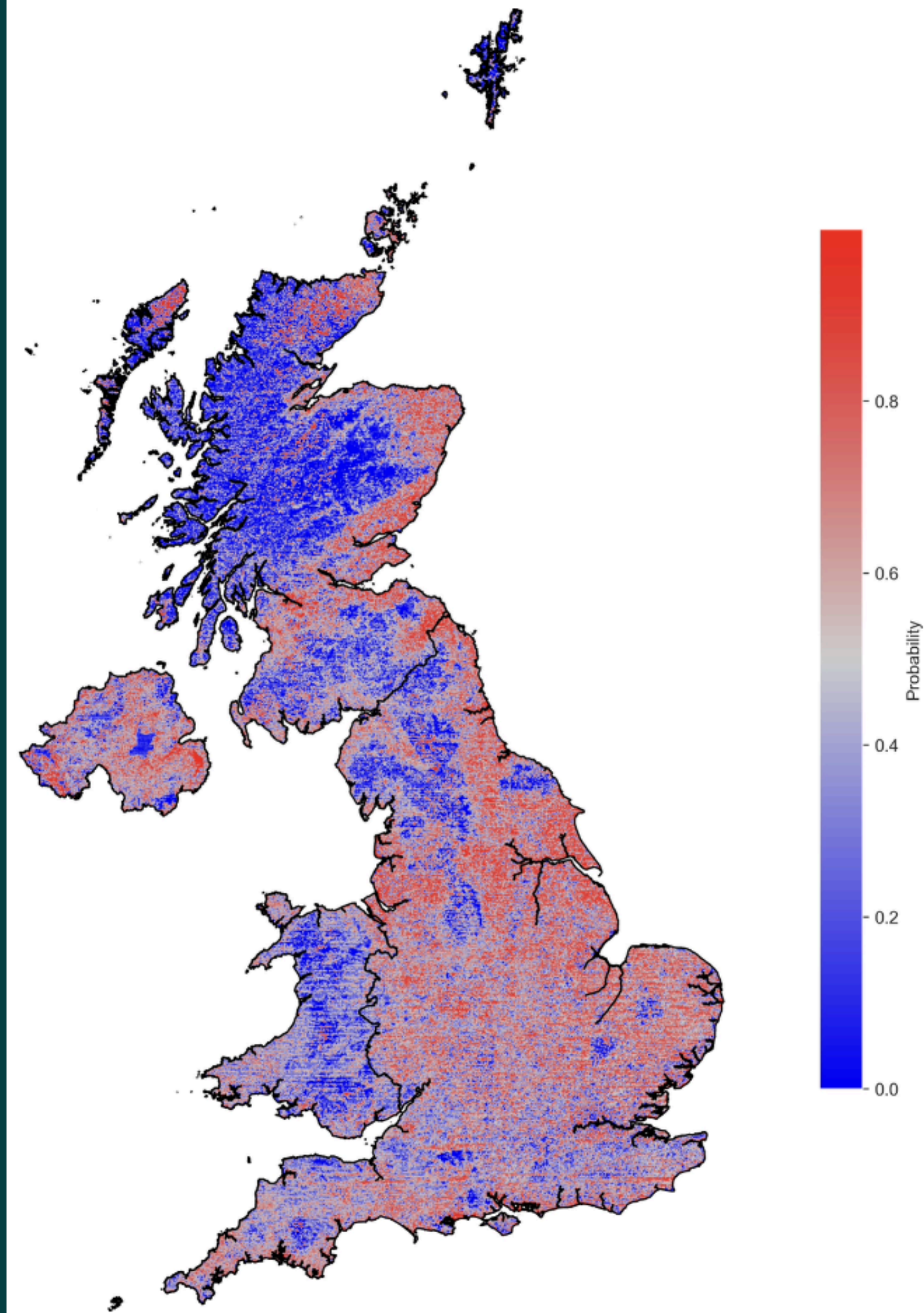
1km² inference grid across UK

Habitat suitability maps
generated per species

Gradient maps (blue = low, red = high)

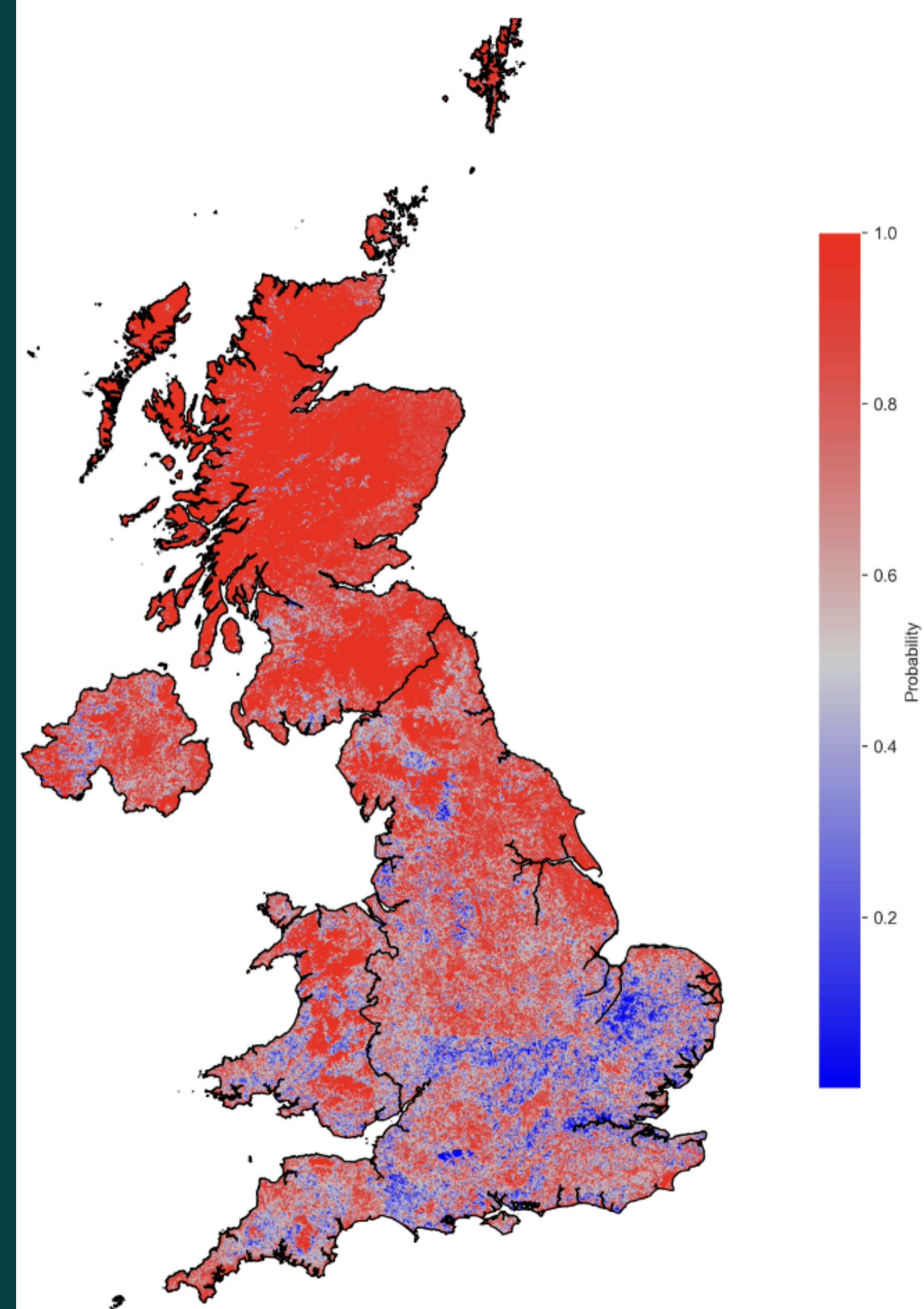
Research Question 1: Suitable Habitats

Hedgehog



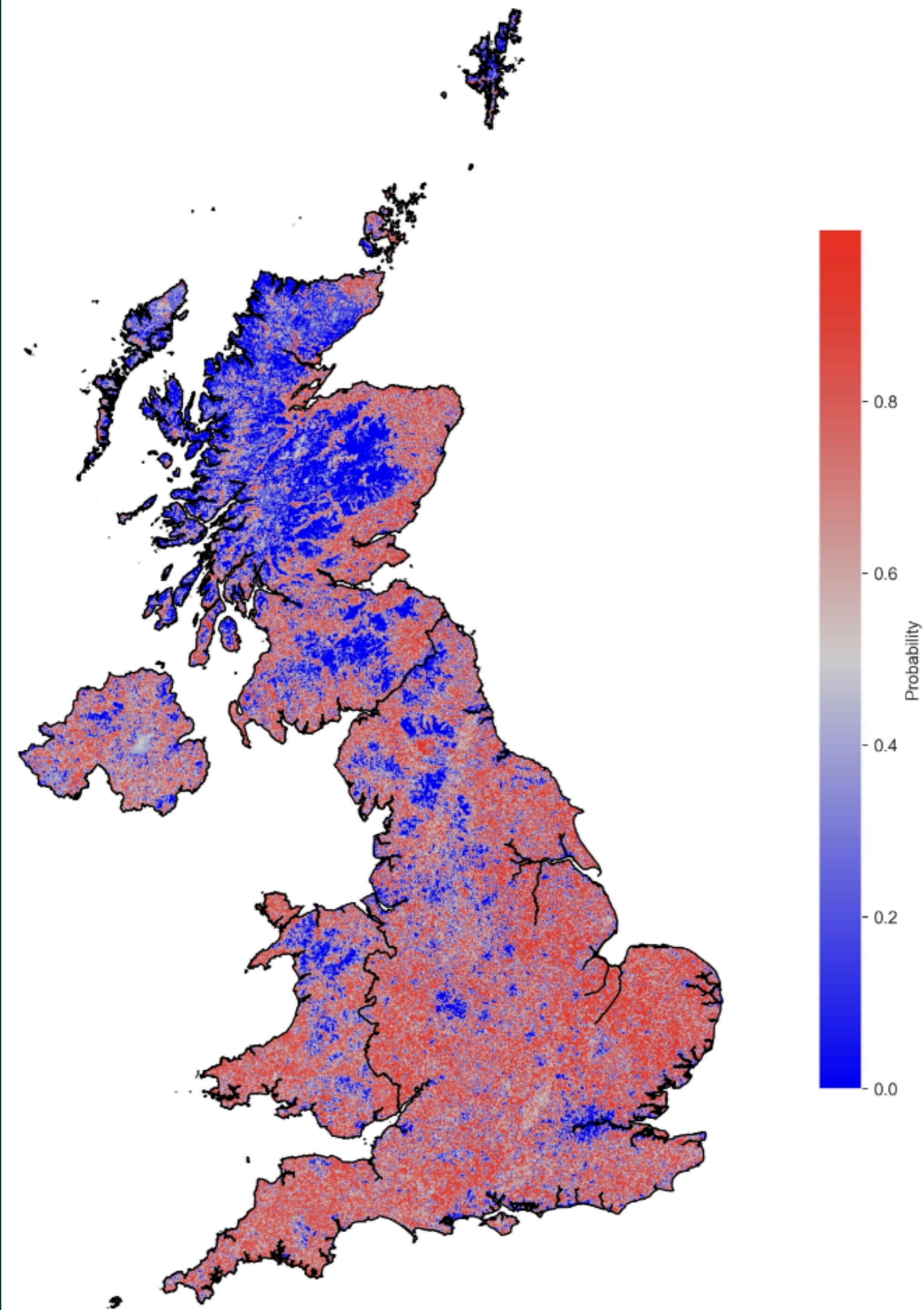
(a) *Erinaceus europaeus* (Hedgehog)

Red Squirrel



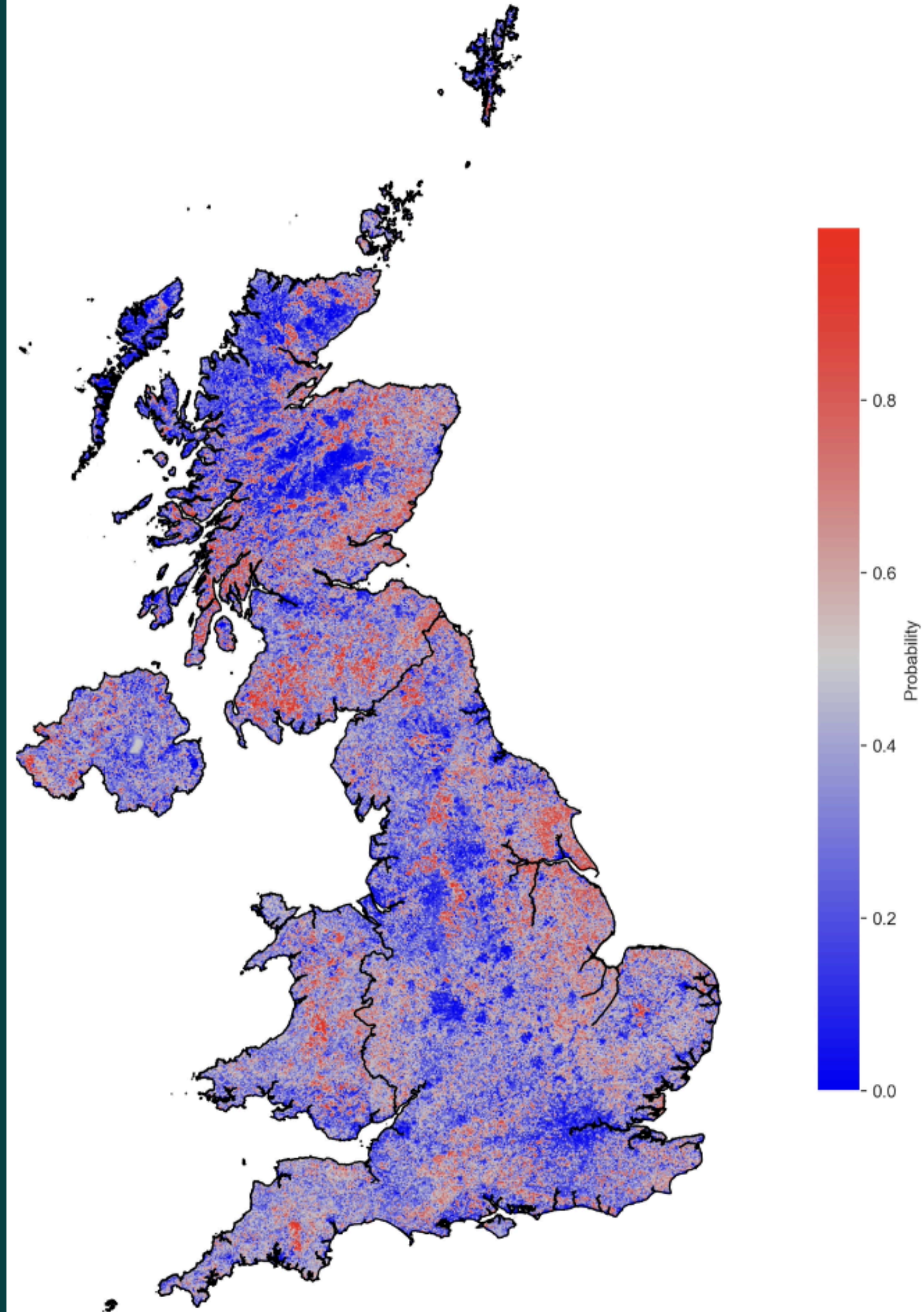
(b) *Sciurus vulgaris* (Red Squirrel)

Hazel Dormouse



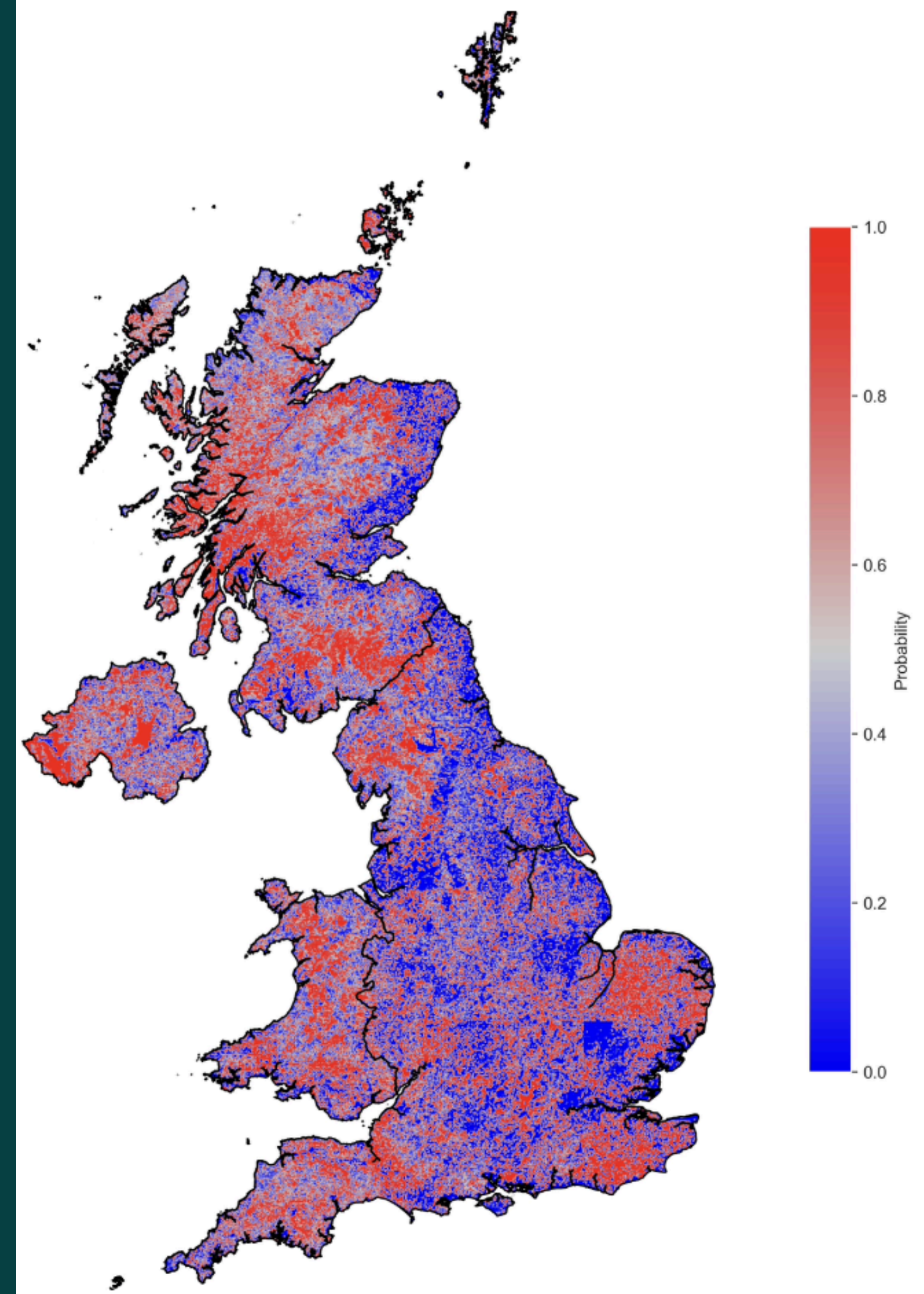
(c) *Muscardinus avellanarius* (Hazel Dormouse)

Brown Hare



(d) *Lepus europaeus* (Brown Hare)

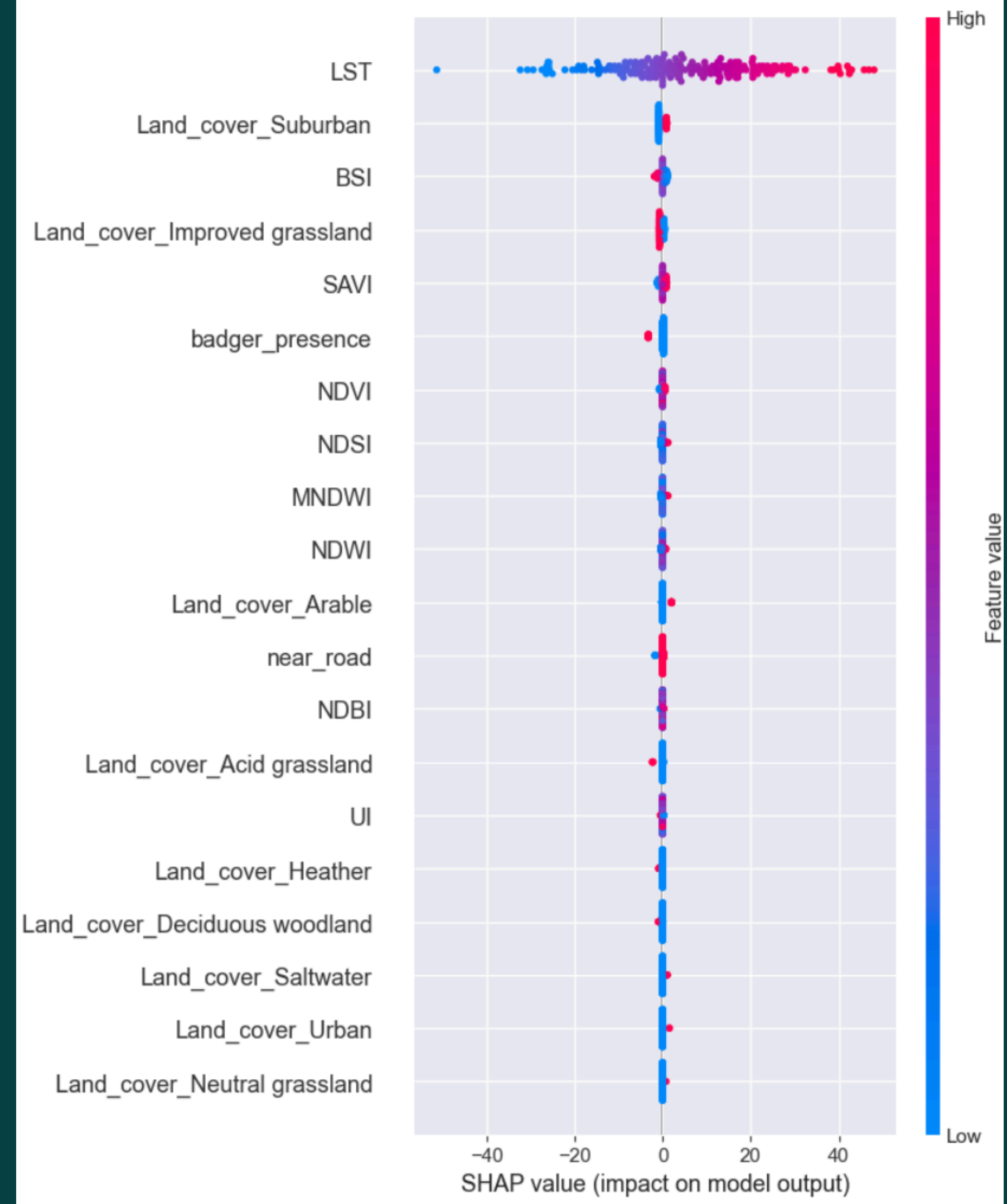
Pipistrelle Bat



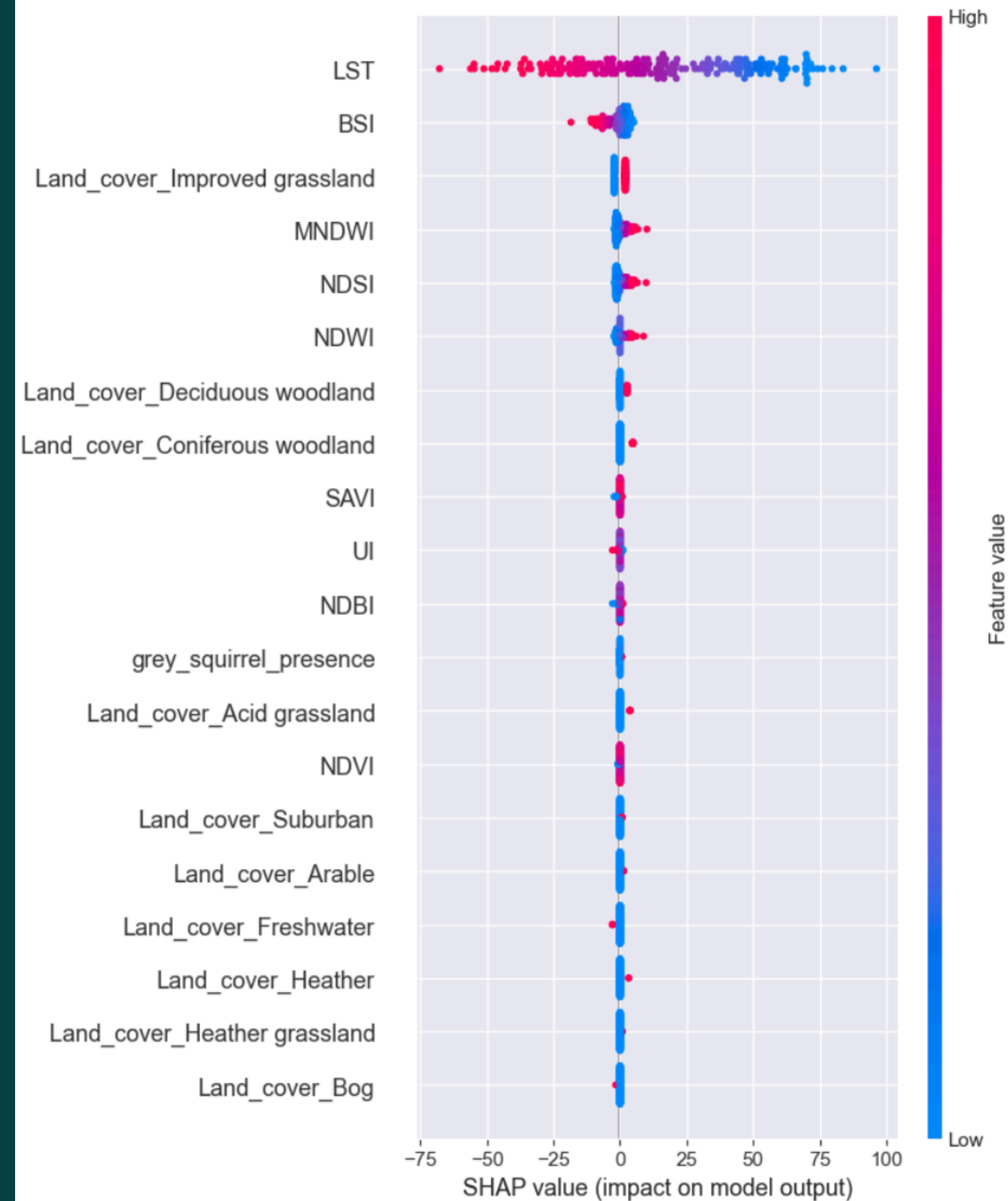
(e) *Pipistrellus pygmaeus* (Soprano Pipistrelle Bat)

Research Question 2: Environmental Drivers

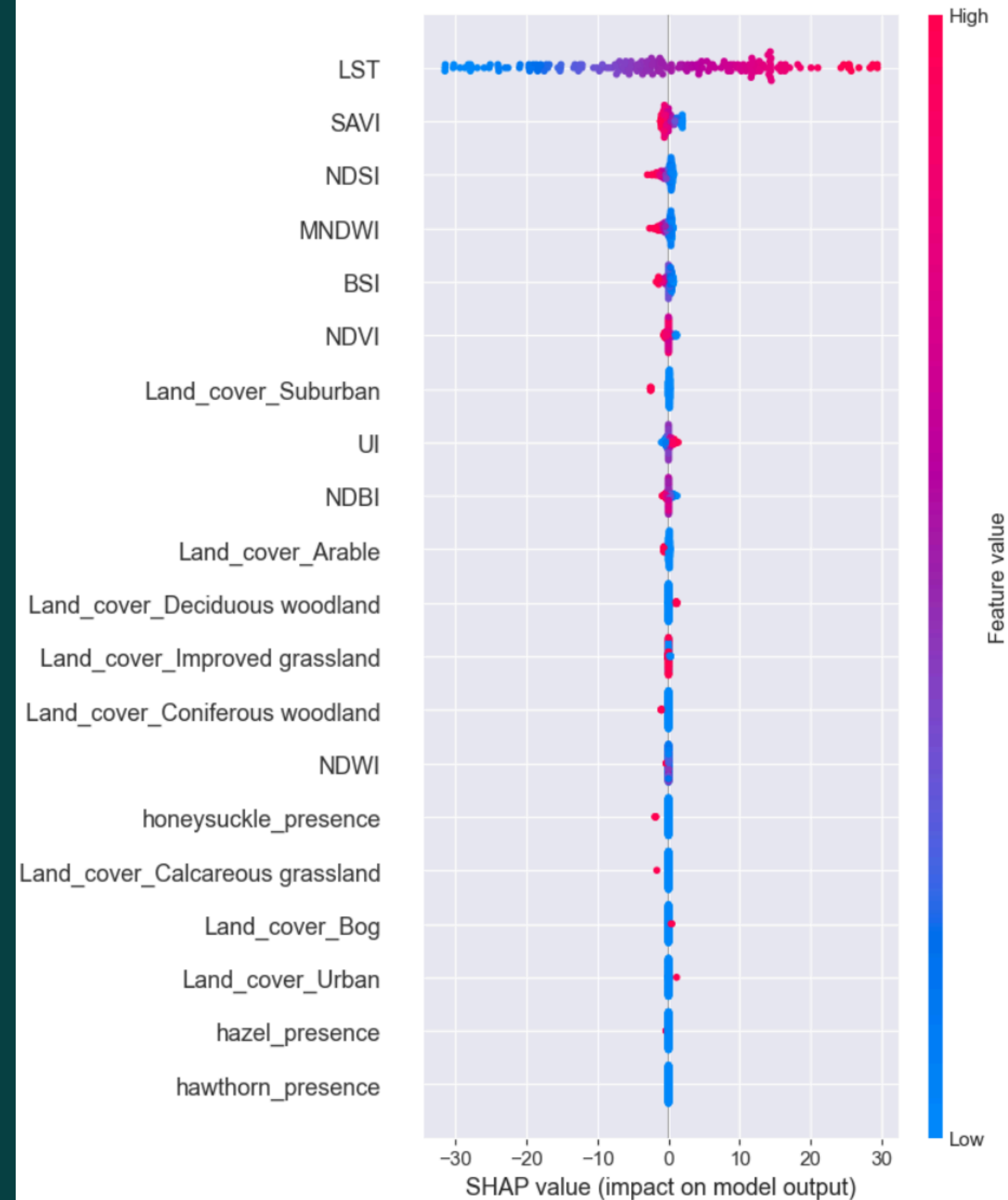
Hedgehog



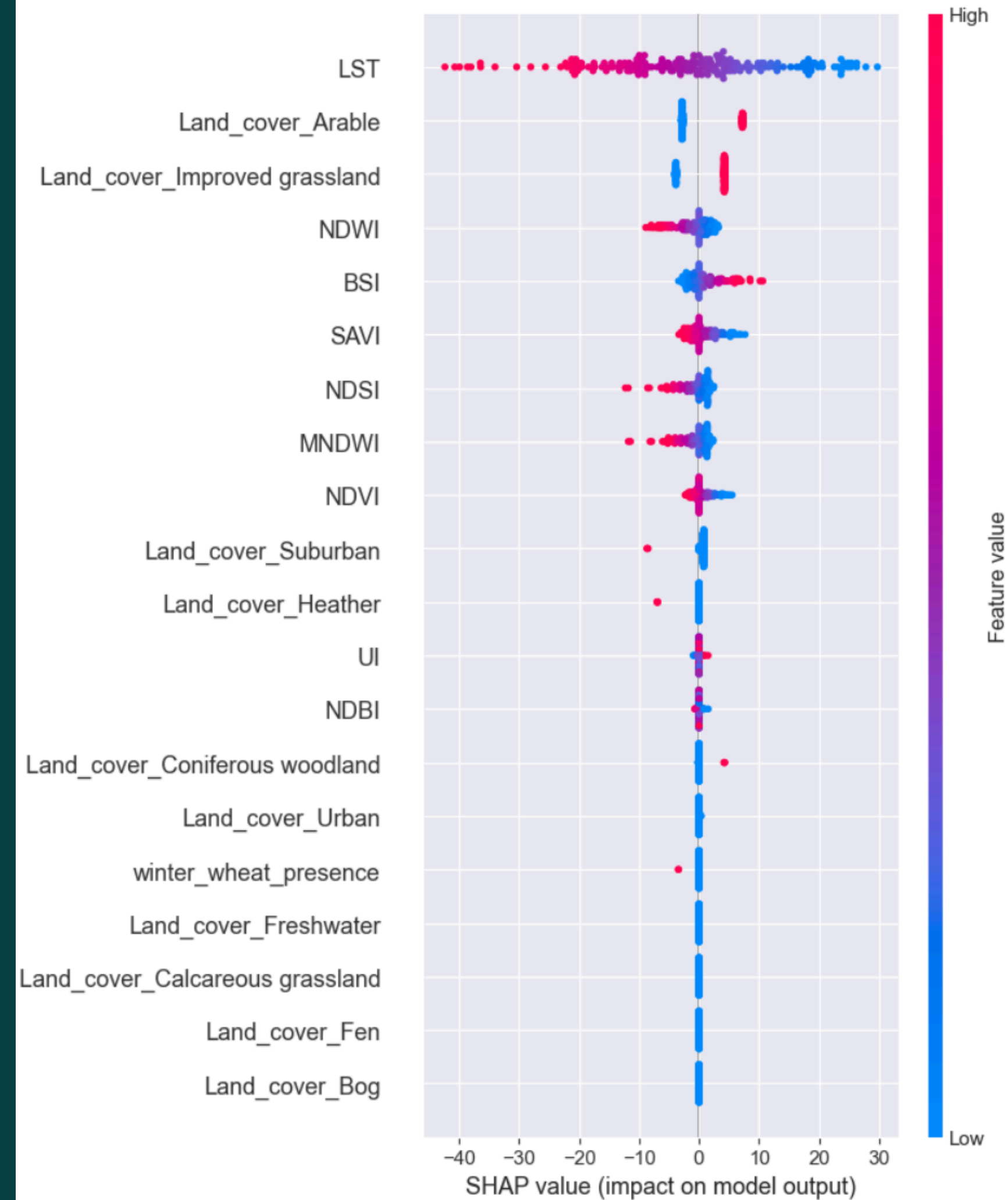
Red Squirrel



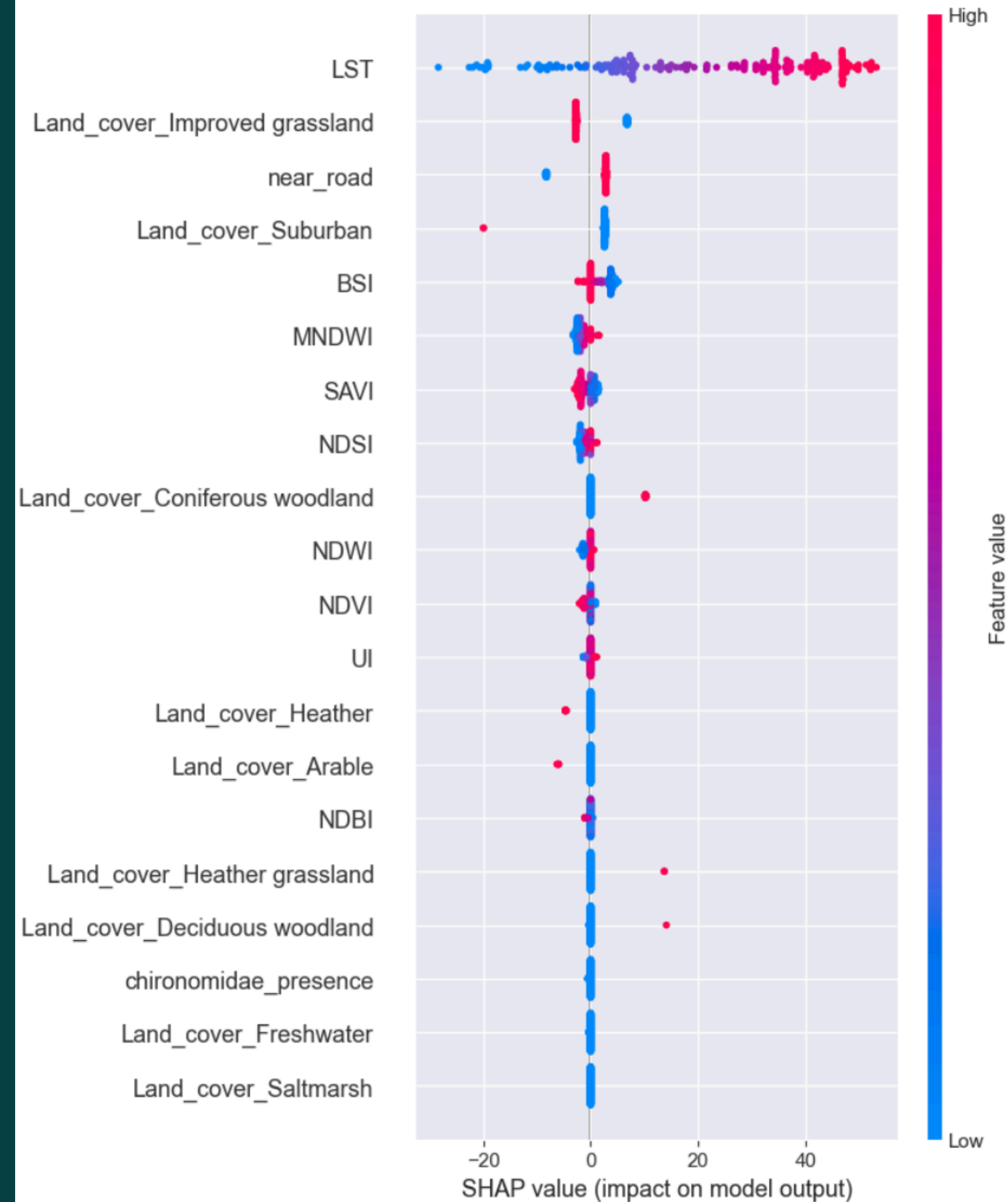
Hazel Dormouse



Brown Hare



Pipistrelle Bat



Research Question 3: Mitigating Data Limitations

Science-to-Policy

- Decision-makers need trusted models for policy use
- Research focused on interpretable decision-support tools, not just predictions
- SHAP analysis used to explain predictions → models become transparent
- Example: Hazel Dormouse suitability linked to dense woodland vegetation
- Habitat maps enable strategic actions:
 - Target monitoring & funding efficiently
 - Justify land management interventions
 - Support habitat protection during planning
- Transparency empowers informed, defensible conservation decisions

Deviations from Plan

- FoP bias assessment dropped (requires true absences, not available)
- Random Forest & XGBoost not used (unsuitable for presence-only data)
- Virtual Ecologist approach considered, then dropped (needs true species distribution)
- Report writing done alongside implementation (caused minor timeline delay but ensured documentation)
- Open-source tool not developed; habitat suitability maps for UK created instead
- Overall, all stages followed sequentially and completed

Topic Selection & Rationale

- **Urban Heat Island Analysis**
 - Pro: Relevant for UK urban planning & climate resilience
 - Con: Satellite thermal resolution too low to separate heat sources
- **Biodiversity Loss Assessment**
 - Pro: Addresses critical conservation issue
 - Con: Too broad; hard to quantify biodiversity via satellite alone
- **Blue Carbon in Seagrass Meadows**
 - Pro: Supports UK seagrass restoration (92% lost)
 - Con: Remote sensing limited by water clarity, depth
- **Invasive Species Monitoring (Japanese Knotweed)**
 - Pro: High economic relevance (£166M annual damage in UK)
 - Con: Difficult to isolate species signature in satellite data
- **Final Choice: Species Distribution Modeling** → focused, achievable, and high conservation relevance

Reflection & Future Work

Reflection, Key Insights:

- Iterative methodology (MaxEnt → DeepMaxEnt) enabled learning & improvement
- Holistic evaluation metrics crucial (e.g., *Pipistrellus pygmaeus* anomaly)
- Data quality > model complexity; DeepMaxEnt succeeded due to cleaned & engineered data
- Ball Tree for proximity analysis → efficient & scalable
- OSMnx library useful for integrating road data
- Data ingestion & cleaning most time-consuming
- Maps intended for ecologists; no formal usability study conducted
- Personal growth: strong Python/data science skills; underestimated data preparation time
- Adaptability was key: dropping unfeasible methods ensured project success

Future Work:

- Field validation with conservation professionals to verify suitability maps
- Seasonal predictions (e.g., winter suitability)
- Include additional variables: population density, habitat connectivity
- GIS integration for planning & protection of endangered species habitats
- Potential SDM testing via species simulation (Garzon-Lopez et al., 2016)

Ethics

Beneficence: Accurate species maps promote well-being of endangered species & UK ecosystems

Non-maleficence: Maps could be misused (poaching, harmful development) → careful controlled sharing

Justice: Results shared with conservationists to maximize public benefit, not commercial gain

Explicability: SHAP analysis used to explain predictions → transparency & trust in AI outputs

Conclusion

- **Objective:** Modeled habitat suitability of 5 endangered UK species using presence-only data
- **Methodology:**
 - Data collection & cleaning (GBIF, GEE, UKCEF)
 - Feature engineering, bias handling, MaxEnt & DeepMaxEnt modeling
 - Ensemble modeling (soft voting) for improved predictions
- **Results:**
 - DeepMaxEnt + ensembles outperformed standard models
 - Habitat suitability maps show species-specific patterns (e.g., *Sciurus vulgaris* in Scotland/N. Ireland, *Muscardinus avellanarius* in southern UK)
 - SHAP analysis revealed key environmental drivers (Land Surface Temperature, land cover, vegetation)
- **Impact:**
 - Provides actionable insights for conservation and policy
 - Demonstrates interpretability of AI models for trusted decision-making
 - Offers a framework for presence-only SDM studies
- **Future Work:**
 - Field validation with conservation professionals
 - Seasonal suitability predictions
 - Include additional variables (human density, habitat connectivity)